

MGIT Result Analysis Portal

Student Academic Result Analysis System I MGIT CSE Department

A full-stack student portal delivering real-time academic insights — grades, GPA trends, and marksheets — built for students, by design.

Presented By: Srinivas Konala – 24261A05R0



Portal Overview

What It Does

The MGIT Student Portal is a secure, role-based academic results platform for the CSE Department. Students can view grades, track GPA trends, and download official marksheets — all from a single dashboard.

Who It Serves

All active and past CSE students across 4 batches, faculty reviewers, and department administrators.

Frontend

React + Vite

Backend

Node.js + Express

Database

MySQL

Security

JWT Auth

STUDENT

ADMIN



MGIT Result Analysis

STUDENT PORTAL

IDENTIFIER

HT Number (e.g. 24261A0501)

PASSWORD

Enter password

SIGN IN →

MGIT CSE • Result Analysis System

Database & Data Scope

1,014

CSE Students

Across 4 active batches (2022–2025)

16

Semester-Batch Combos

Covering all active semester-batch pairings

~35K

Result Rows

Cleaned and verified examination records

 Only **regular examinations** are included. Supplementary/re-sit results are deliberately excluded to maintain GPA calculation consistency. All data has been cleaned and verified before ingestion.



Authentication & Security

The portal uses a stateless, token-based authentication model designed for scalability and security.

→ **JWT-Based Login**

Signed tokens issued at login — no server-side session storage required.

→ **bcrypt Password Hashing**

All passwords are hashed with bcrypt before storage — never stored in plaintext.

→ **Role-Based Access Control**

Students can only access their own records. Admin roles unlock broader data views.

Dashboard

LATEST SEMESTER - SEM 3

SGPA

9.52

Semester 3

CGPA

9.19

Overall

PASSED

9

subjects this sem

BATCH RANK

#28

of 318 · Year 2



Dashboard Tab

The landing tab gives each student an instant academic snapshot — no navigation required.

Latest SGPA

Current semester's grade point average, computed from live result data.

Cumulative CGPA

Running CGPA across all completed semesters.

Pass / Fail Count

Quick count of passed and failed subjects in the latest semester.

Batch Rank

Latest overall rank across all the students present in a year-batch.

Results Tab

The Results Tab is the core browsing interface for academic records. Results are grouped by semester for clarity, with flexible filtering tools to help students locate any subject instantly.

Grade Badge Color Codes

O — Outstanding

B+ — Good

A+ — Excellent

B — Average

A — Very Good

C — Pass

F — Fail (highlighted in red)

Semester Dropdown

Filter the entire results view to a specific semester.

Subject Search Bar

Live search by subject name or code across all semesters.

Grouped Display

Results organized by semester with clean table rows per subject.

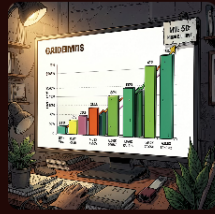
GPA Overview Tab

A comprehensive academic performance dashboard — rankings, trends, and cumulative progress visualized in one view.



Rank Cards

Batch rank and section rank cards computed against peers in real time.



SGPA Bar Chart

Semester-by-semester SGPA shown as a proportional bar trend for quick pattern recognition.



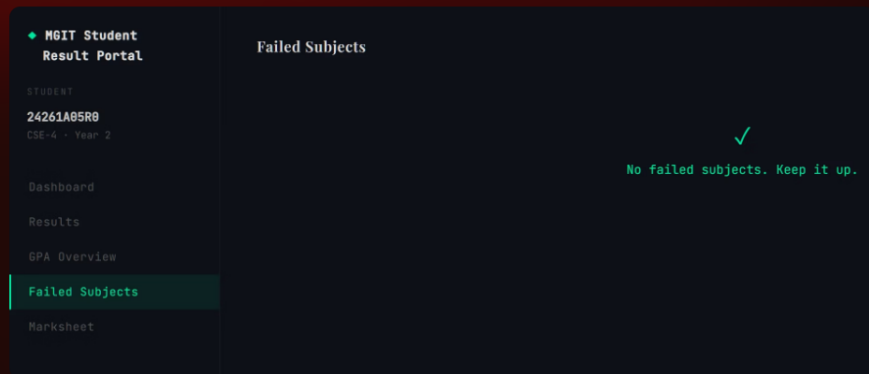
CGPA Line Chart

Cumulative GPA plotted across all semesters, showing academic trajectory over time.



CGPA Card

Prominent display of the student's overall cumulative GPA, updated after each semester.



Failed Subjects Tab

A dedicated view to help students track and address academic backlogs — without digging through every semester manually.



All-Semester Aggregation

Every failed subject across all semesters is surfaced in one clean list.



Semester Filter

Optional dropdown to narrow failures to a specific semester.



Empty State

When no failures exist, a clean "no failures found" state is shown — a reward for consistent performance.

Marksheet Tab

Students can generate and download official semester marksheets as PDFs — produced on demand directly from live database records, ensuring accuracy at the moment of generation.

Per-Semester PDFs

One marksheet generated per semester — select the term, download instantly.

Generated On Demand

No pre-stored files. PDFs are rendered fresh from live data on every request.

Always Accurate

Because data is sourced live, marksheets always reflect the current verified record — no stale exports.

Key Technical Decisions

1

JWT Over Sessions

Stateless tokens eliminate server-side session storage, enabling horizontal scalability and simpler deployment across environments.

2

HT Number as Identity Source

Batch, year, and section are derived directly from the student's Hall Ticket number — no redundant data entry, no sync issues.

3

Excluding Supplementary Results

Only regular exam results are used for GPA computation. Including supplementary scores would skew semester averages and misrepresent academic standing.

4

Proportional SGPA Bars

SGPA trend bars are scaled proportionally (not fixed-height), giving an honest visual representation of performance variation across semesters.

The background of the image is a dark red, almost black, fabric that is draped and folded in a way that creates deep shadows and bright highlights, giving it a three-dimensional, flowing appearance. The folds are organic and fluid, with some areas appearing more saturated than others.

**Thank You for your
attention to this matter!**